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Machine learning prediction of bio-oil characteristics quantitatively relating to biomass compositions and pyrolysis conditions

Tonghuan Zhang [a], Danyang Cao [a], Xin Feng [a], Jiahua Zhu [a], Xiaohua Lu [a], Liwen Mu [a,*], Hongliang Qian [b,*]

[a] State Key Laboratory of Materials-Oriented Chemical Engineering, Nanjing Tech University, Nanjing 211816, PR China
[b] Department of Pharmaceutical Engineering, China Pharmaceutical University, Nanjing 211198, PR China

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ABSTRACT

It is crucial to predict the characteristics of pyrolytic bio-oil accurately for its application, but the prediction results are greatly influenced by biomass compositions and pyrolysis conditions. In this work, different biomass compositions analysis (chemical compositions, ultimate and proximate analysis) and pyrolysis conditions (particle size, heating rate and pyrolysis temperature) were successfully used as input to analyze the characteristics of bio-oil by machine learning method. The model based on ultimate analysis is better for regression analysis of the yield, viscosity and oxygen-carbon ratio (O/C) of bio-oil. The model based on chemical compositions is better for regression analysis of calorific value and hydrogen-carbon ratio (H/C) of bio-oil. Moreover, relative error analysis and scatter diagrams were used to analyze the predicted results. In addition, the analysis of partial dependence diagram shows the influence of various factors and the interactions on the target variables. This study provides feasible thinking for the prediction of the characteristics of bio-oil obtained by biomass with different compositions under different pyrolysis conditions.

1. Introduction

The rapid consumption of non-renewable fossil energy has caused serious problems of energy security and environmental pollution [1–3], and considerable attention has been devoted to biomass energy due to its abundance and renewable in nature [4–6]. Thermochemical conversion is capable of converting biomass into biofuels using an efficient and economical way [7,8], which are then synthesized into the desired chemicals or used directly [9]. The main thermochemical conversion of biomass are combustion, gasification and pyrolysis [10], in which pyrolysis, a thermal decomposition process under oxygen free conditions, is considered as a potential biomass conversion method due to the relatively easy operation and speed [7,11].

Bio-oil is one of complex products of biomass pyrolysis and has been sparked huge interests from academia and industry [12–14], which can be used as boiler fuel, transportation fuel and chemical products [15]. There are more than 350 compounds in bio-oil, including acids, aldehydes, ketones, esters, alkanes and various aromatic compounds [16], which are mainly composed of C, H and O elements. H/C and O/C can be used as indexes of bio-oil quality [10]. For example, the hydrogen contents of bio-oil usually reflect the calorific value and chemical structure [17]. Besides, there are some characters demand of the bio-oil such as high calorific value (CV), suitable viscosity and stability during its application [18], which could make bio-oil compatible with the current energy infrastructure [19]. For example, viscosity is an important characteristic for the gas turbine fuel affects its pumping and atomization [20]. The dynamic viscosity of bio-oil is 0.0604–0.8319 Pa·s at 20 ℃, which is larger than that of commercial diesel oil (0.0012–0.0052 Pa·s) with good atomization [20]. In addition, bio-oil has a lower CV because of its high-water content and more oxygenated compounds as compared with transportation fuels, which requires further upgrading treatment [21].

The characteristics of bio-oil can be determined by the diverse biomass resources (such as forest residues, agricultural residues [22]) and different pyrolysis conditions (such as biomass size, pyrolysis temperature and heating rate [22]). On the one hand, the diversity of biomass compositions is reflected in the complex structure of different biomass resources. From the smallest scale (angstrom, 10^{-10} m), the chemical bonds of biomass are composed of carbon, hydrogen, oxygen and nitrogen; chemical bonds can also form biopolymers (10^{-10} to 10^{-9} m), and biomass compositions can be regarded as cellulose, hemicellulose and lignin [23]. The three components eventually constitute

* Corresponding authors.

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biomass, which spans multiple orders of magnitude in scale [24]. In addition, biomass compositions can be expressed as fixed carbon, ash and volatiles in proximate analysis. On the other hand, biomass compositions interact with each other, and the difference of compositions content leads to the difference of the components of pyrolysis bio-oil [25,26]. For example, the hydrocarbon content decreases linearly and ketone content increases with the increasing cellulose content, respectively. The furan content increases with the increase of hemicellulose content and phenol content increases linearly with increase of lignin contents [27]. In addition, the biomass compositions and pyrolysis conditions change the distribution of products and affect the characteristics of bio-oil [28]. For example, the hazelnut feed pellets of 0.425–0.6 mm could produce the maximum amount of bio-oil via the fixed bed pyrolysis [29]. Ozbay et al. reported the yield of bio-oil from cottonseed cake at 300 ℃/min was 32.1% higher than that at 5 ℃/ min in a fixed bed pyrolysis process [30]. Therefore, the relationship between the characteristics of bio-oil and biomass compositions and pyrolysis conditions is unclear due to the complex reaction for the biomass conversion, which also is the difficulty and hotspot of research. In addition, this is different from the relationship between the characteristics of bio-oil and their own compositions in literature.

A great deal of work has been done to establish kinetic models by thermogravimetry and scanning calorimetry to study the relationship between the characteristics (only yield and compositions) of bio-oil and biomass compositions and pyrolysis conditions [31,32]. Noritatsu et al. established the kinetic model of biomass pyrolysis by simulating the design of biomass pyrolysis in fluidized bed. The calculated data are in good agreement with the experimental data of pine sawdust, lignin and cellulose pyrolysis at very high temperature (800 ℃) [33]. Unfortunately, the kinetic study of one certain biomass and its corresponding pyrolysis condition is not universal, and the obtained model cannot be applied to other biomass systems directly. Therefore, it is particularly important to obtain the universal relationship between the bio-oil's characteristics and biomass compositions and pyrolysis conditions.

With the development of artificial intelligence, many new methods have been introduced into traditional research, and satisfactory results have been obtained [34–38]. There are many machine learning methods such as random forest, support vector machines (SVM) and gradient boosting (GB) that can be used to predict the parameters. Among these methods, random forest (RF) is a commonly used and effective algorithm [39,40]. Random Forest is an integrated research method based on tree predictor, which can solve regression and classification problems simultaneously. Zhou et al. successfully studied the effects of catalysts and solvents on biomass hydrothermal liquefaction yield and bioenergy performance by random forest method [39]. Zhu et al. established the correlation between bio-char yield, biomass structure information and pyrolysis conditions, and successfully used RF method to accurately predict the yield of bio-char [40]. Xing et al. used the RF model in the final analysis to accurately predict the chemical compositions of various biomasses and compare it with previous relevant data and experimental data [41]. Tang et al. predicted the production and hydrogen content of bio-oil based on the compositions of biomass feedstock and pyrolysis conditions, indicating that RFs is more feasible than MLR to predict the production and hydrogen content of bio-oil [17]. There are many comprehensive studies on the internal relationship between bio-oil and high-quality production, however, most of the studies focus on the effects of operating conditions from an experimental point of view. There are rare studies focus on the effects of operating conditions and the compositions of biomass itself simultaneously, especially from the perspective of machine learning [42].

In this work, a comprehensive analysis of the pyrolytic bio-oil, including quantity (yield), quality (viscosity and calorific value) and compositions (H/C and O/C) prediction was presented. Meanwhile, the feasibility of predicting viscosity, calorific value, H/C and O/C of bio-oil by machine learning was verified. Using machine learning tools, the yield, viscosity, calorific value, H/C and O/C of bio-oil were predicted according to biomass compositions and pyrolysis conditions, and the correlation analysis was carried out. This study provides feasible thinking for the prediction of the characteristics of bio-oil obtained by biomass with different compositions under different pyrolysis conditions.

2. Methods

2.1. Data collection

The experimental data of bio-oil yield (282 datasets), viscosity (101 datasets), calorific value (198 datasets), H/C (214 datasets), and O/C (232 datasets) were collected from literature and listed in Table S1 of the Supporting Information. Each data contained valid values of all variables. For all biomass samples, the raw materials are usually ground into smaller pieces, and then processed by chemical structure characterization, ultimate and proximate analysis. Under inert conditions, biomass with a certain particle size (PS) was pyrolyzed at a certain heating rate (HT) and pyrolysis temperature (PT). In addition, there are inevitably experimental errors in the collected data: the accuracy of data was gained at different temperatures and units, which meant the viscosity may contain the errors of different densities. Unity mainly depends on equation (1):

$$\mu = \nu\rho/1000 \tag{1}$$

where μ and ν indicate the dynamic viscosity (Pa·s) and kinematic viscosity (mm^2/s), and ρ is the density (g/cm^3).

2.2. Data pretreatment

The biomass compositions and pyrolysis conditions are the important factors to affect the characteristics of the pyrolytic bio-oil. The critical features of biomass compositions from biomass to bio-oil are categorized into three groups: (i) the chemical components of raw materials, including cellulose, hemi-cellulose and lignin's relative contents (Cel-Hem-Lig); (ii) ultimate analysis of raw materials, including the element contents (C–H-O-N); (iii) proximate analysis of raw materials, including the matter contents of volatiles, ash and fixed carbon (VM-Ash-FC).

The biomass compositions, pyrolysis conditions and targets (yield, viscosity, calorific value, H/C and O/C of bio-oil) are statistically described by using semi-boxplot. After data statistics are completed, Pearson correlation coefficient (calculated according to formula (2)) is used to describe the linear correlation of any two variables statistically.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\sum_{i=1}^{n}(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2}$$

where $\bar{x}$ or $\bar{y}$ denotes the mean values of variable x or y. The significance level of the correlation coefficient of any two variables is tested by equation (3):

$$t = \frac{r\sqrt{N-2}}{\sqrt{1-r^2}} \tag{3}$$

where N is the number of samples, r is the Pearson correlation coefficient (PCC). Hence, p-value is gained according to two-tailed t distribution with N-2 degrees of freedom. In this work, PCC is utilized to detect the collinearity among any two input variables and the linear relationship between input and target variables [43]. PCC value is between [-1, 1], which can simply judge the linear correlation between data. The larger absolute value means the better linear relationship between variables. Positive value indicates the positive correlation, and vice versa. Therefore, PCC can be used to analyze the influence trend of characteristic variables on target variables.

2.3. Random forest model

Random forest algorithm is a kind of ensemble learning algorithm, which is a double random bagging model, behaving data and feature selection randomness. A certain number of data and feature attributes are randomly selected in the training data set, and decision tree model is generated by training in base learner of RF model. A forest formed via many of these decision tree models could be integrated into the prediction results, which reduces the prediction deviation and improves the prediction accuracy.

2.3.1. Random forest model principle

Bagging, also known as bootstrap aggregation method, is a typical representative of parallel algorithms in ensemble learning. In essence, Bagging is an integration technique that forms several new data sets based on the original data set by adopting put-back sampling. Multiple different data sets obtained by sampling are trained and learned to obtain the base learning model, and then a strong learning model is formed by combining these multiple base learning models. The output value of the strong learner finally obtained by Bagging is based on the form of several weak learners [42].

The Bagging algorithm flows are as follows.

(1) Input a training sample set $D = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots (x_m, y_m)$, and its data size is m.
(2) Generate T new training data sets, each data set is sampled back for m times, and a sampling training set D_t ($t = 1,2, \dots T$) containing m samples is obtained.
(3) Generate T base learner models based on T sampling training sets with set base learners.
(4) The final strong learner is based on T base learner models [43].

2.3.2. Random forest regression

Normalize the values of the data set using formula (4), and then input the model to start data analysis.

$$x_i^* = (x_i - \mu)/\sigma \tag{4}$$

where x_i^* is the normalized value of input variable and x_i is the

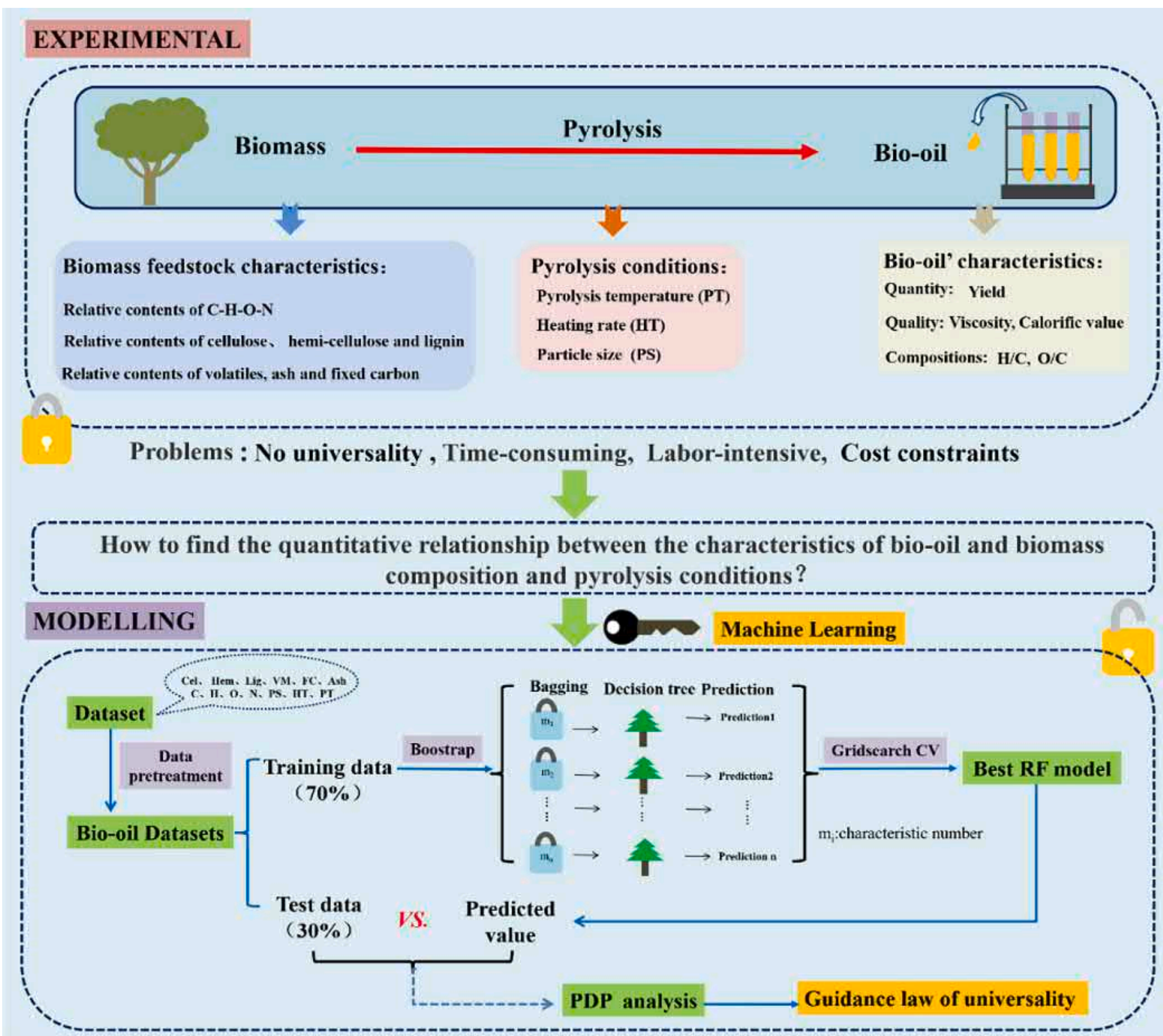

Fig. 1. Research framework of the prediction of yield, viscosity, calorific value, H/C and O/C of bio-oil based on biomass compositions and pyrolysis conditions via machine learning method.

original value of input variable. μ and σ are the mean value and standard deviation of each input variable, respectively.

In this work, different biomass compositions and pyrolysis conditions as feature information were input for the training and prediction, the best feature combination (R^2 value is large, RMSE value is small) was selected as the model for subsequent analysis. RF is selected and used to train and optimize the models of yield, viscosity, calorific value, H/C and O/C prediction using the scikit-learn Python library (version 0.19.2), respectively. Python code related to data preprocessing training was provided in Table S2 in supplementary information. There are three steps for the utilization of RF [44]: (i) randomly sampling the data set with return to form a training set; (ii) The decision tree is trained by different subsamples, and the decision of each node of the decision tree is determined based on randomly selected features; (iii) The best splitting mode is calculated according to randomly selected features, and each tree-based model is integrated into a random forest model. The data is divided into training and test set with a ratio of 70:30 (Fig. 1). Moreover, the hyper-parameters are resolved via a five-fold gross-validation grid-search method. According to the specific training model, the model parameters are selectively adjusted according to the discrepancy between the data sets such as the number and the maximum depth of trees. In addition, the minimum number of samples for dividing internal nodes and leaf nodes remain the default values (i.e., 2 and 1, respectively). The hyper-parameter is then used to retrain the model, and the remaining 30% data was used to test.

2.3.3. Evaluation indicators

Regression coefficient (R^2) and root mean square error (RMSE) are often utilized as evaluation indexes in regression analysis, so the performance of the developed model is tested by equations (5) and (6).

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(Y_i^{\mathrm{exp}} - Y_i^{\mathrm{pred}}\right)^2}{\sum_{i=1}^{N}\left(Y_i^{\mathrm{exp}} - \overline{Y}^{\mathrm{exp}}\right)^2} \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Y_i^{\mathrm{exp}} - Y_i^{\mathrm{pred}}\right)^2} \tag{6}$$

where Y_i^{exp} and Y_i^{pred} are the actual and predicted values, respectively. $\overline{Y}^{\mathrm{exp}}$ is the mean of the actual values.

In order to distinguish the prediction accuracies, four models are developed with various input variables: (i) all input variables; (ii) pyrolysis conditions and C–H-O-N; (iii) pyrolysis conditions and Cel-Hem-Lig; (iv) pyrolysis conditions and VM-FC-Ash, respectively. Meanwhile, the error analysis and visual representation of the predicted results and actual values are carried out by using the scatter plot. And RF superior model could be used to understand the effect of each feature through the

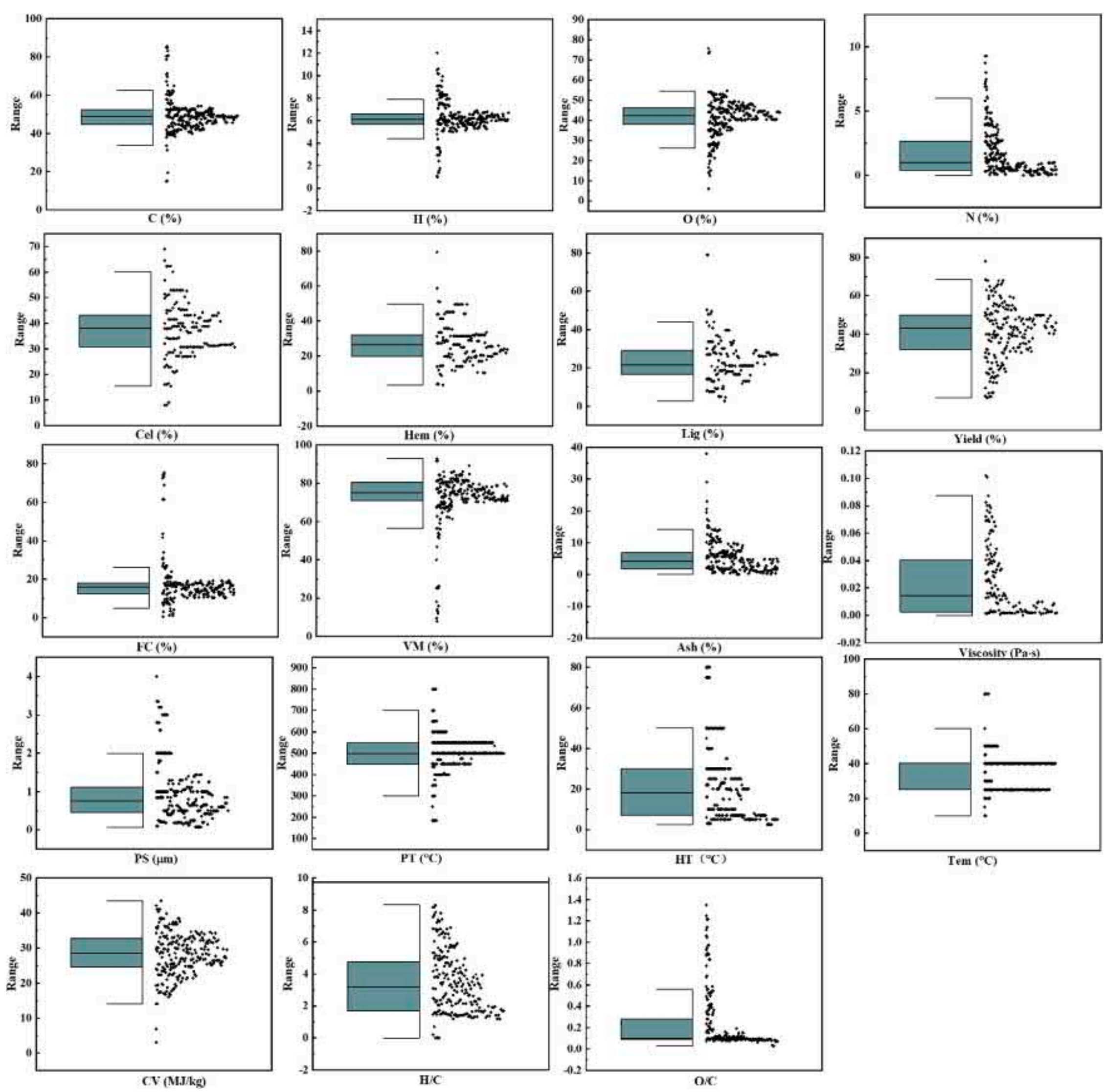

Fig. 2. Box plot of statistical variables related to biomass dataset.

mean decrease impurity method. In addition, in order to study the influence of input characteristic variables on the yield, viscosity, calorific value, H/C and O/C of bio-oil, partial dependence plot (PDP) was used for univariate analysis. At the same time, the complex relationship between the interaction of the two features and the target variable is further analyzed by using the two-factor PDP.

3. Results and discussion

3.1. Dataset description and statistical analysis

According to the ultimate analysis, chemical compositions, and proximate analysis of the feedstock, the statistical features of the biomass are obtained in Fig. 2 and Tables S3-S5 in the Supporting Information. Data description is given on the left side of the semi-box chart, including upper limit, upper quartile, median, lower quartile and lower limit, while specific data points are drawn on the right side of the semi-box chart as shown in Fig. 2. The ultimate C, H, O and N element analysis of the biomass are in the range of 14.97–71.5%, 3.21–12.03%, 16.59–74.01%, and 0.08–9.29%, respectively. Among them, the content of C, H and O are relatively concentrated in 50%, 6% and 42%, while the content of N is relatively low. The chemical components data (cellulose, hemicelluloses and lignin) are in the range of 8.00–69.00%, 4.00–49.66%, and 7.61–79%, respectively. In addition, the contents of chemical components in the biomass are relatively dispersed. The fixed carbon, volatile matter, and ash contents of the biomass dataset are in the range of 0.50–43.60% 40.00–86.04%, and 0.10–19.50%, respectively. At the same time, the contents of fixed carbon, volatile matter and ash are about 18%, 75% and 4%, respectively. The ranges of compositions analysis exhibit divergence, which may be attributed to high diversity of biomass resources. The pyrolysis conditions (i.e., PS-HT-PT) of the biomass dataset are in the range of 0.08–3.20 mm (PS), 3–100 ℃/min (HT), and 300–800 ℃ (PT), respectively. The calorific value and yield are in accordance with normal distribution, which also reflects the diversity of bio-oil preparation results under different biomass and pyrolysis conditions. Bio-oil behaves

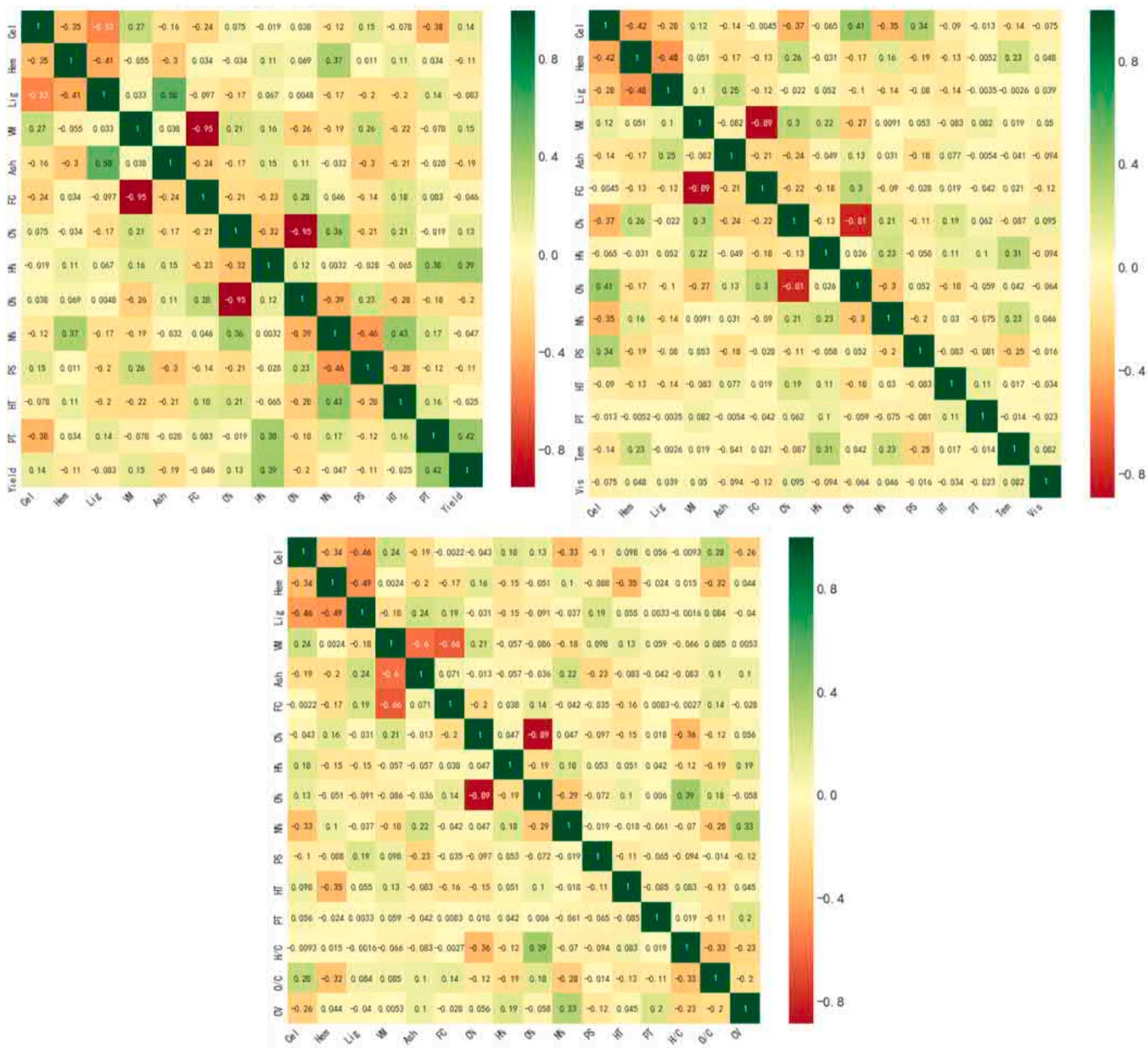

Fig. 3. Pearson correlation matrix among any two variables.

the high viscosity and O/C, and low H/C, which are reflected in the statistics.

PCCs between any two variables are shown in Fig. 3, and the correlation coefficient is reported in Tables S6-S8 in the Supporting Information. The value of p represents the reliability of describing the linear correlation. There are remarkably linear correlations in the following variables: PT-yield ($p < 0.01$), hydrogen contents-yield ($p < 0.01$), nitrogen contents-viscosity ($p < 0.01$), FC-calorific value ($p < 0.01$), oxygen contents-calorific value ($p < 0.01$), nitrogen contents-calorific value ($p < 0.01$), carbon contents-H/C ($p < 0.01$), oxygen contents-H/

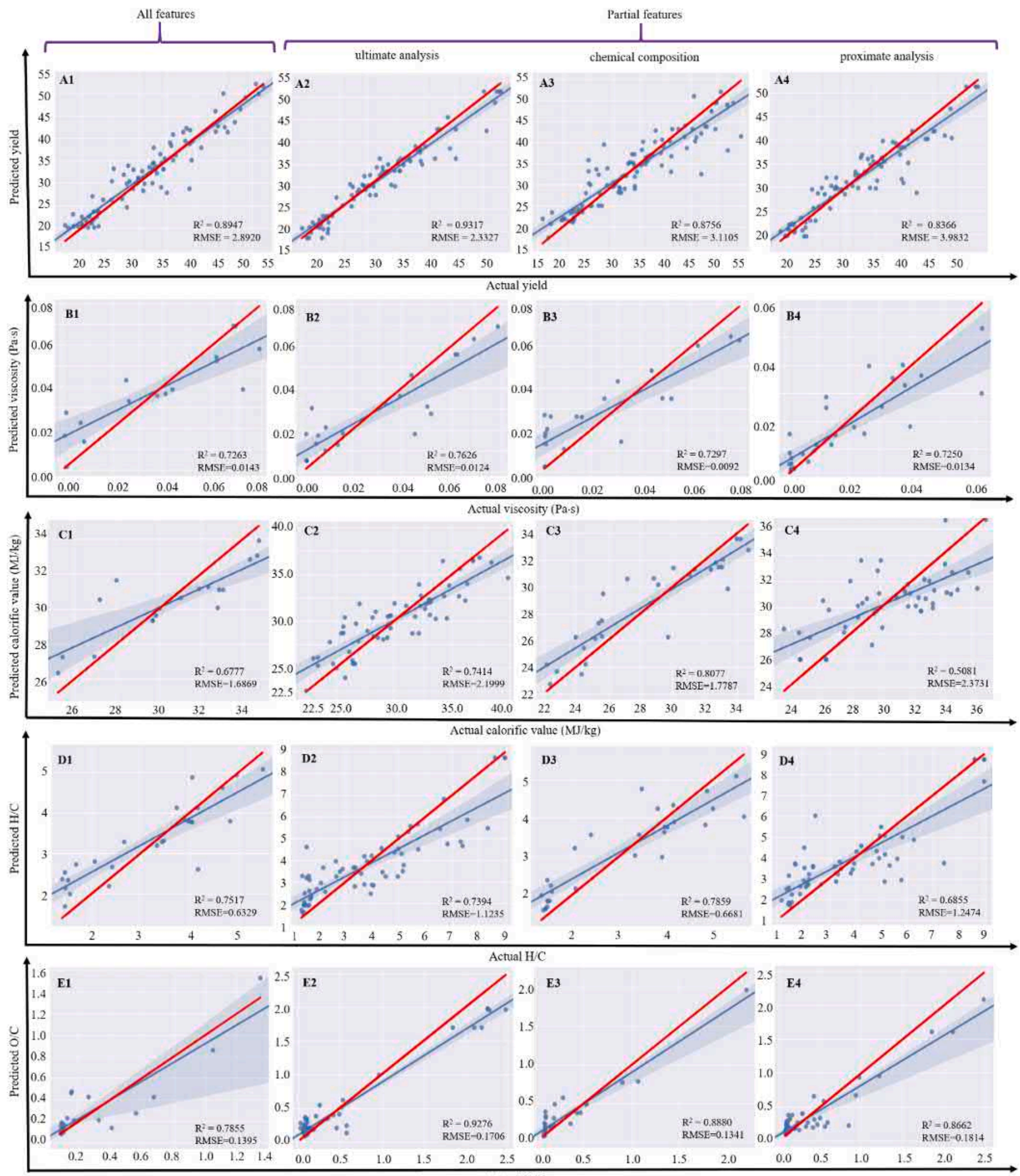

Fig. 4. Comparison of predicted and actual values with different inputs: A-E represents different target variables, from left to right: all inputs, pyrolysis conditions and C–H-O-N, pyrolysis conditions and Cel-Hem-Lig, pyrolysis conditions and VM-FC-Ash, respectively. Red lines refer to the predicted values = actual values line.

C ($p < 0.01$), hemicellulose-O/C ($p < 0.01$), carbon contents-O/C ($p < 0.01$), oxygen contents-O/C ($p < 0.01$). The PT and hydrogen contents have positive effects on the yield of bio-oil, with Pearson correlation coefficients of 0.42 and 0.39, respectively. Biomass pyrolysis needs heat, and higher pyrolysis temperature in a certain range is beneficial to break bonds in biomass to obtain larger yield. The nitrogen content of bio-oil is positively correlated with viscosity, and the correlation coefficient is 0.046. This may be related to the relative atomic mass of nitrogen atoms, and the viscosity of bio-oil increases with the increasing molecular weight [45]. The CV of bio-oil is negatively correlated with FC content and O content, and positively correlated with N content, with correlation coefficients of 0.028, 0.058 and 0.33 respectively. This phenomenon can be explained as follows: the higher the FC content is, the lower yield and CV of bio-oil is according to the law of mass and energy conservation. The compositions of bio-oil is closely affected by the original biomass compositions, so the CV of bio-oil is negatively correlated with oxygen content and positively correlated with nitrogen content [27]. Based on the close correlation between bio-oil compositions and biomass compositions, H/C is obviously negatively correlated with C content in bio-oil. Meanwhile, the O/C of bio-oil is positively correlated with hemicellulose, and the correlation coefficient is 0.015, which may be due to the easy pyrolysis of hemicellulose and less hydrocarbon compounds. FC claims a significantly linear relationship with both VM ($p < 0.01$) and ash ($p < 0.01$). Moreover, there is no obvious linear relationship among the yield, viscosity, calorific value, H/C, O/C and other variables of bio-oil. Among them, PCCs shows that there is no significant correlation among most variables. Therefore, we use a more detailed RF model as an interpretable model with the best performance, and the performance of the model obtained by using different biomass analysis descriptions as feature variables will be discussed in the following sections.

3.2. Evaluation of developed RF model

The super parameters of random forest algorithms are selected through five cross-validation, and are used to retrain the prediction models for the yield, viscosity, calorific value, H/C and O/C of bio-oil with various input variables in Fig. 4, respectively. The 30% experimental data that are used to value the performances of the developed RF models. The red and blue line represent $y = x$ and regression line, respectively. All the RF optimal models reach an acceptable level that R^2 scores are larger than 0.75 and RMSE scores are close to 0. It is indicated that the model based on ultimate analysis has better performance for estimating the yield, viscosity and O/C of bio-oil in Fig. 4 (A1-A4), (B1-B4) and (E1-E4), respectively. While the chemical compositions model has good ability to predict the CV and H/C of bio-oil in Fig. 4 (C1-C4) and (D1-D4), respectively.

The prediction results of the RF model are illustrated by visualizing the prediction values and actual data in Fig. 5. It is shown that the range of yield, viscosity, CV, H/C and O/C are located around 15–55%, 0.001–0.08 Pa·s, 22–36 MJ/kg, 1.0–6.0 and 1.1–2.7, respectively. In addition, we have compared the prediction results with actual data, which is indicated that the RF model is suitable to predict the yield, viscosity, calorific value, H/C and O/C of bio-oil. In a word, the accuracy of the final target analysis model is acceptable. For example, the R^2 obtained by using the model based on ultimate analysis is 0.93, which is better than the bio-oil yield predicted by using random forest in the past ($R^2 = 0.92$) [17].

Fig. 6 shows relative error between the prediction results using existing RF models and actual data, in which the red dotted lines represent the data distribution with a relative error of 20%. It is worth noting that the most relative error values are within 20%. The overall relative error of yield and CV of bio-oil was relatively small, which is related to the normal distribution. The relative error distributions of viscosity, H/C and O/C are relatively messy, which can be caused by the complexity of bio-oil components.

3.3. Feature importance

Fig. 7 evaluates the relative importance of the input features of the optimal performance model to yield, viscosity, O/C (RF model based on ultimate analysis), CV, H/C (RF model based on chemical compositions) and the absolute value calculated by PCCs. PCCs refers to the linear correlation between the target variable and each variable, and the relative importance includes linear or nonlinear correlation based on RF model [46]. The both Pearson and RF model show that the hydrogen contents of feedstock primarily affect the yield and viscosity of bio-oil, the cellulose contents of feedstock govern CV, the PS of feedstock affects H/C, and HT has the direct impact on O/C. There is no obvious relationship between the importance of features described by RF model

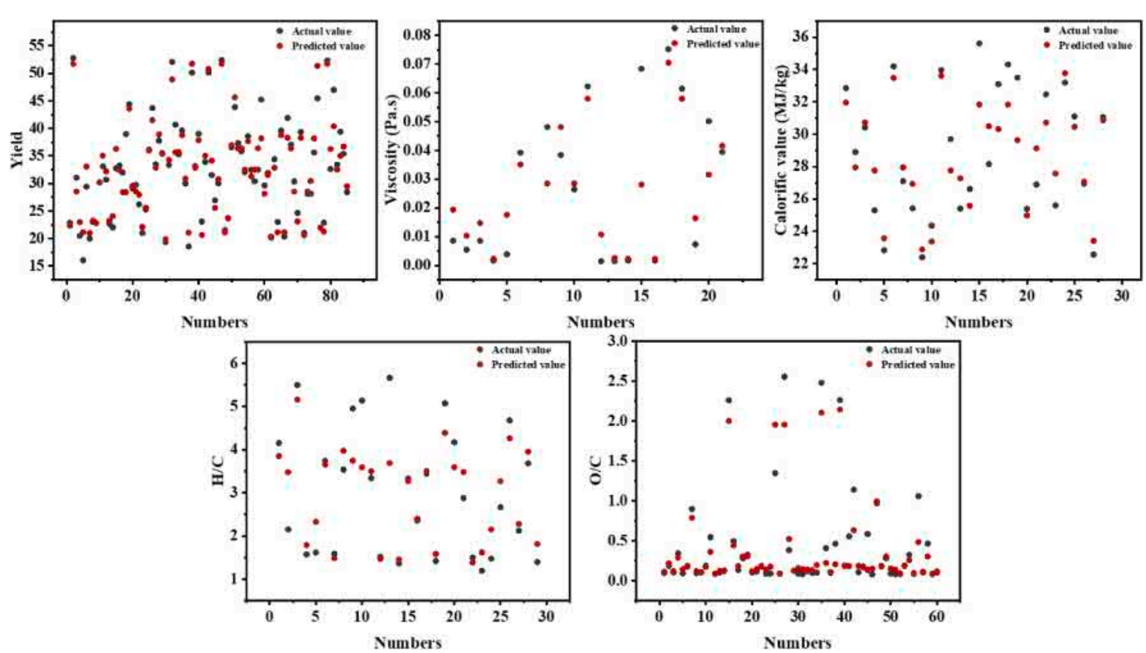

Fig. 5. The correlation between biomass sets achieved from actual value and predicted value from trained RF.

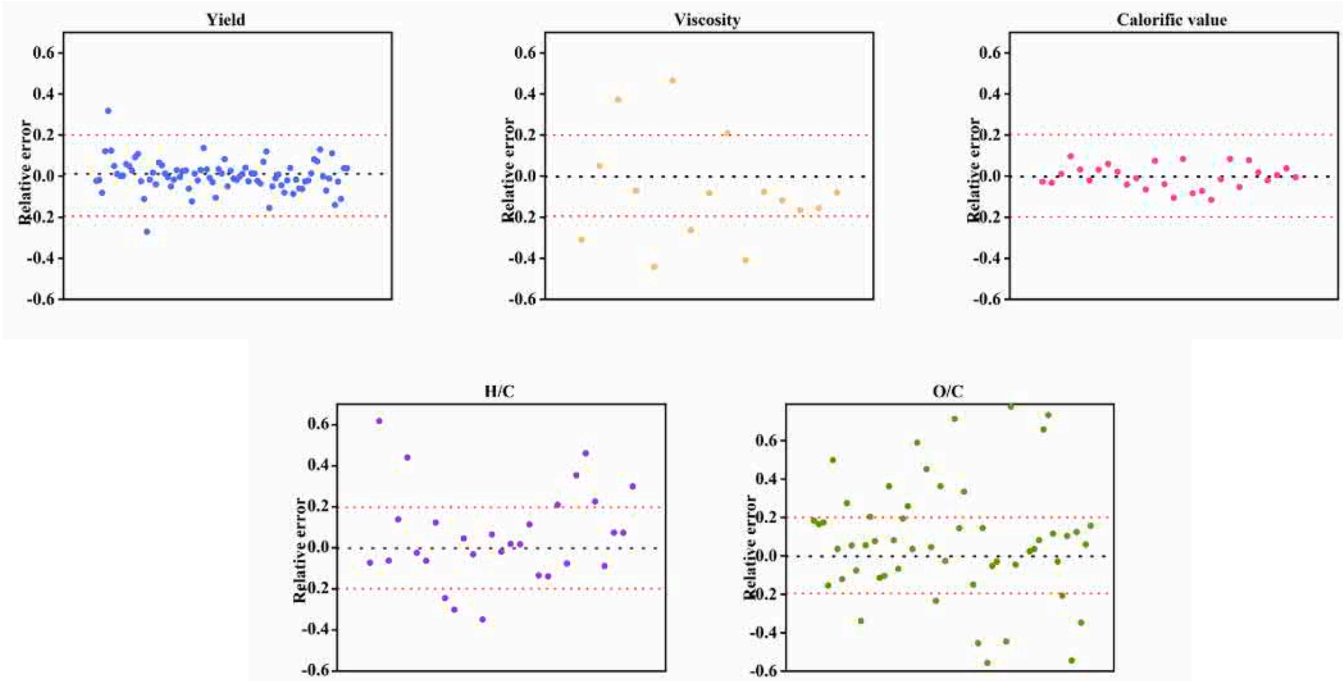

Fig. 6. RF models prediction performance for the yield (based on ultimate analysis data), viscosity (based on ultimate analysis data), calorific value (based on chemical compositions data), H/C (based on chemical compositions data) and O/C (based on ultimate analysis data) of bio-oil.

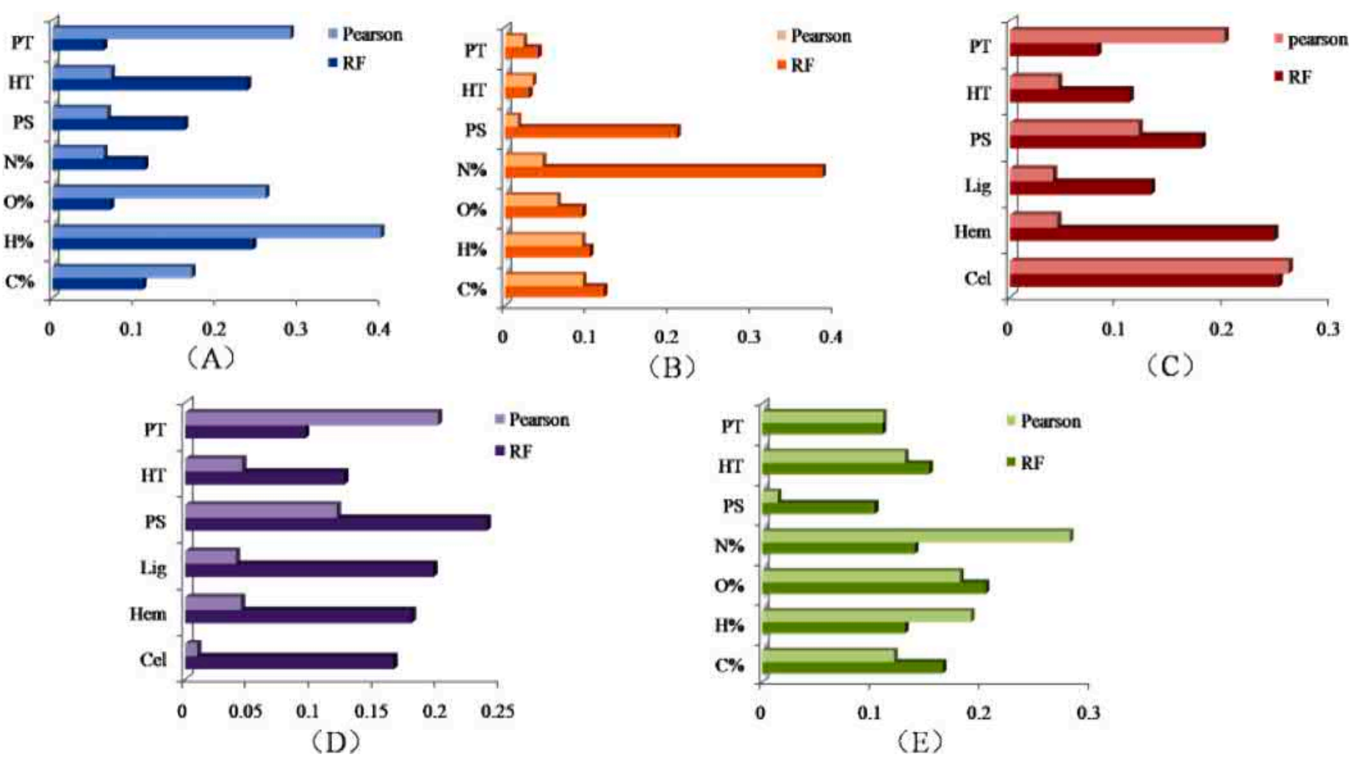

Fig. 7. Feature importance of input variables via RF and PCCs: (A) yield, (B) viscosity, (C) calorific value, (D) H/C and (E) O/C.

and the linear correlation between features described by Pearson analysis and targets, which confirms that bio-oil yield, viscosity, calorific value, H/C and O/C are not direct linear relationships with the input variables.

It is indicated from the RF model that the feedstock compositions have the greatest influence. In other words, the ultimate analysis data and the chemical compositions of raw materials are more important than pyrolysis conditions for characteristics of bio-oil, which is similar as the literature results [17]. As far as pyrolysis conditions are concerned, the results of PCCs and RF model are inconsistent. The features arranged in descending order of importance based on the RF model are as follows: HT, PS and PT, which may be due to the relatively uneven distribution of experimental data HT, PS and more uniform and concentrated distribution of PT. Furthermore, the RF model based on ultimate analysis data shows that N and O are still the greatest influential factors on the viscosity and O/C value, which accounted for the 24% changes in the viscosity and the 20% changes in the O/C of bio-oil, respectively. Meanwhile, the RF model based on chemical compositions shows that cellulose, hemicelluloses and PS are still the greatest influential factors, in which the first two features accounted for the 25% changes in the calorific value and the last one accounted for the 24% changes in the H/C of bio-oil.

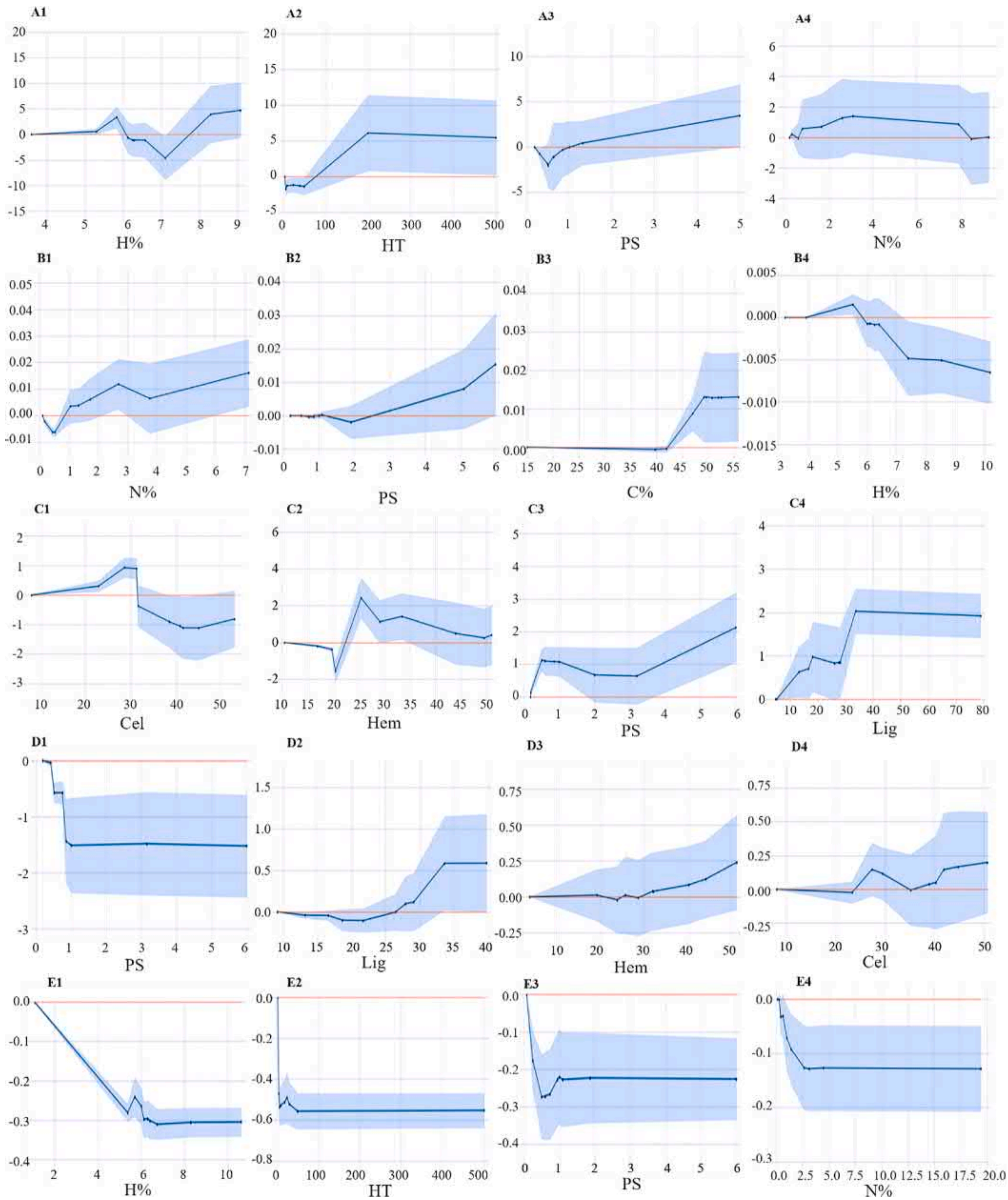

Fig. 8. The partial dependence plot of A-E represents yield, viscosity, calorific value, H/C and O/C vs some variables. The peak on the X axis represents the quantile of the objective eigenvalue, which reflects the data density.

3.4. Analysis of univariate partial dependence plots for correlation predictions of bio-oil

The features importance of the target variable is obtained by the optimal model, and the partial correlation analysis is carried out on the important feature variables of yield, viscosity, calorific value, H/C and O/C of bio-oil. By PDP analysis, the impact trends of features on target variables can be obtained. This method is used to adjust one feature to predict the target variables (i.e., yield, viscosity, calorific value, H/C and O/C of bio-oil), while the other features are limited to mean values. It is noted that when data points are sparse, trend lines may be inaccurate and ignored. The blue line plots are drawn by prediction data, instead of scatter points of original actual data. The partial dependence of different feature variables on the target are shown in Fig. 8.

The yield of bio-oil decreases when the hydrogen contents in the feedstock increase in range of 6–7%, and thereafter increase with the increase of hydrogen contents (Fig. 8A1). In addition, the yield of bio-oil is positively correlated with nitrogen contents in the range of 1–3%, and then decreases with the increasing nitrogen contents (Fig. 8A4). There is the positive correlation between HT and yield, and the slope of this correlation decrease with the increasing HT (Fig. 8A2), which is verified in literature [47]. At high HT, the short time of tar cracking and repolymerization in secondary reaction is the main reason for the enhancement in the yield of bio-oil. The smaller the PS (<0.5 mm) of biomass is, the higher the bio-oil yield is. With the increasing PS (greater than0.5 mm), the yield of bio-oil increases (Fig. 8A3), which was consistent with the results described in literature [48].

With the increase of nitrogen and carbon content in biomass raw materials, the viscosity of bio-oil increases accordingly (Fig. 8B1 and B3), which is negatively correlated with hydrogen content (Fig. 8B4). This can be explained by the molecular weight of C, N is larger than that of H, and larger molecular weight could lead to higher viscosity [45]. There is a positive correlation between PS and viscosity of bio-oil (Fig. 8B2). The small PS will easily degrade, resulting in low molecular weight of the component and the low viscosity.

Different structural contents of biomass components promote the interaction between components to change accordingly, thus affecting the distribution of bio-oil components. With the growing cellulose and hemi-cellulose contents in biomass feedstock, the CV of bio-oil increases firstly and then decreases (Fig. 8C1 and C2), which means that a suitable content of cellulose and hemicellulose is beneficial for the enhancement in the CV of bio-oil [49]. When lignin content in biomass feedstock is in the range of 10–40%, the CV of bio-oil shows an upward trend generally in Fig. 8C4. However, the lignin content is unduly high, the yield reduces and the water content raises, which could be due to the formation of lignin derivatives [48]. When PS is in the range of 0.5–3 mm, the CV of bio-oil decreases with the increase of PS in Fig. 8C3, which is mainly attributed to the lower water content of raw materials with smaller PS. This will lead to higher carbon and hydrogen content of bio-oil [50].

It is shown that PS is negatively correlated with the H/C of bio-oil in Fig. 8D1, mainly because the components of bio-oil are governed by the original compositions of biomass [27]. The small original particle will result in the small water content of biomass and large H/C of bio-oil. As for the chemical compositions of biomass, it is indicated that the H/C of bio-oil increases monotonically with the increase of lignin and hemicellulose content (Fig. 8D2 and D3), which may be attributed to the oxygenated compounds and a small amount of hydrocarbons in bio-oil [51]. The main sources of oxygen content in bio-oil are carboxyl and carbonyl based chemicals gained from cellulose pyrolysis, and phenol methoxy based chemicals are also obtained from lignin pyrolysis [52]. However, the H/C of bio-oil decreases when the cellulose content increases between 25% and 35%, and then increases with the further increase of cellulose content (Fig. 8D4). Bio-oil with hydrocarbon compositions is mainly produced by cellulose degradation, so the higher cellulose content will lead to the greater H/C [53].

The components of bio-oil is highly correlated with the original compositions of biomass [45]. The O/C of bio-oil increases with the increase of hydrogen content range from 5 to 5.8% and reduces with the further increase of hydrogen content (Fig. 8E1). Nitrogen content has negatively correlated with O/C of bio-oil (Fig. 8E4). The O/C of bio-oil gradually increases with the increase of HT (≤30 ℃/min), and then decreases with the further increase of HT (Fig. 8E2). At high HT, the generation of oxygen-containing gas such as carbon dioxide or CO will increase, resulting in the decrease of oxygen content, which is consistent with the experimental results in the literature [54]. The effect of PS on the O/C is as follows: the minimum value of O/C appears when PS was 0.5 μm, and then shows an upward trend with the increase of PS (Fig. 8E3). This because the larger PS will lead to the slower the heat transfer rate [55].

3.5. The two-factor partial dependence plots analysis for bio-oil correlation predictions

In order to further analyze and predict the relationships between complex input features and bio-oil yield, viscosity, calorific value, H/C, O/C targets, PDP with two-factor can show more complex relationships between targets and features. It should be noted that the features we want to analyze are independent of other features. The yield is more dependent on hydrogen content at lower pyrolysis temperature (Fig. 9A1), because the compositions of bio-oil components depend on hydrogen. When the pyrolysis temperature is higher than 200 ℃, the yield has little correlation with nitrogen content and particle size (Fig. 9A2 and A3). When the PS is small enough, the bio-oil yield is more dependent on hydrogen content (Fig. 9A4), which is consistent with the previous machine learning results [17].

It is shown that the viscosity is closely related to the original element compositions of raw materials. At a small PS (<1 mm), the viscosity of bio-oil has nothing to do with carbon content (Fig. 9B1), but little to do with hydrogen content (Fig. 9B2), and is more dependent on nitrogen content (Fig. 9B3). With the increase of biomass nitrogen content, the viscosity of bio-oil increases. At the same time, when the hydrogen content of biomass is<6%, there is a significant positive correlation between the viscosity of bio-oil and the PS of raw materials. Meanwhile, the elemental content of biomass also restricts the viscosity of bio-oil. When the hydrogen content of raw materials is higher than 8%, the viscosity has nothing to do with the carbon content of biomass. At the relatively low hydrogen content, the viscosity of bio-oil is negatively correlated with the carbon content of biomass itself (Fig. 9B4). When the content of hydrogen and carbon in biomass is relatively small, they have great effect on the viscosity.

The results show that the CV is closely related to the original PS of biomass. When the PS is smaller than 6 mm, the CV decreases with the increase of cellulose content (Fig. 9C1), and the correlation with hemicellulose is mainly reflected in the medium hemicellulose content (20–40%) (Fig. 9C2), but almost has nothing to do with lignin content (Fig. 9C3). There is a correlation between biomass compositions and pyrolysis temperature. When hemicellulose content is constant, the CV increases with the increasing pyrolysis temperature (Fig. 9C4), mainly because the increasing PT promotes the pyrolysis of coke, and then improves the gas quality [56].

It is showed that H/C of bio-oil is closely related to biomass PS. When the biomass PS is larger than 2 mm, the H/C has little correlation with the content of cellulose and hemicellulose (Fig. 9D1 and D2). In addition, when the content of cellulose and hemicellulose is constant, the H/ increases with the reduction in particle size. There is a synergistic effect between biomass PS and lignin content. When the PS is smaller than 2 mm, the H/C is significantly correlated with the increase of lignin content. When the PS is larger than 2 mm, the H/C increases with the increasing lignin content (Fig. 9D3). Especially, the chemical structure of biomass itself also significantly affects the H/C, and lignin content has a more significant effect on the H/C than cellulose content. When the cellulose content is constant, the H/C increases with the increasing

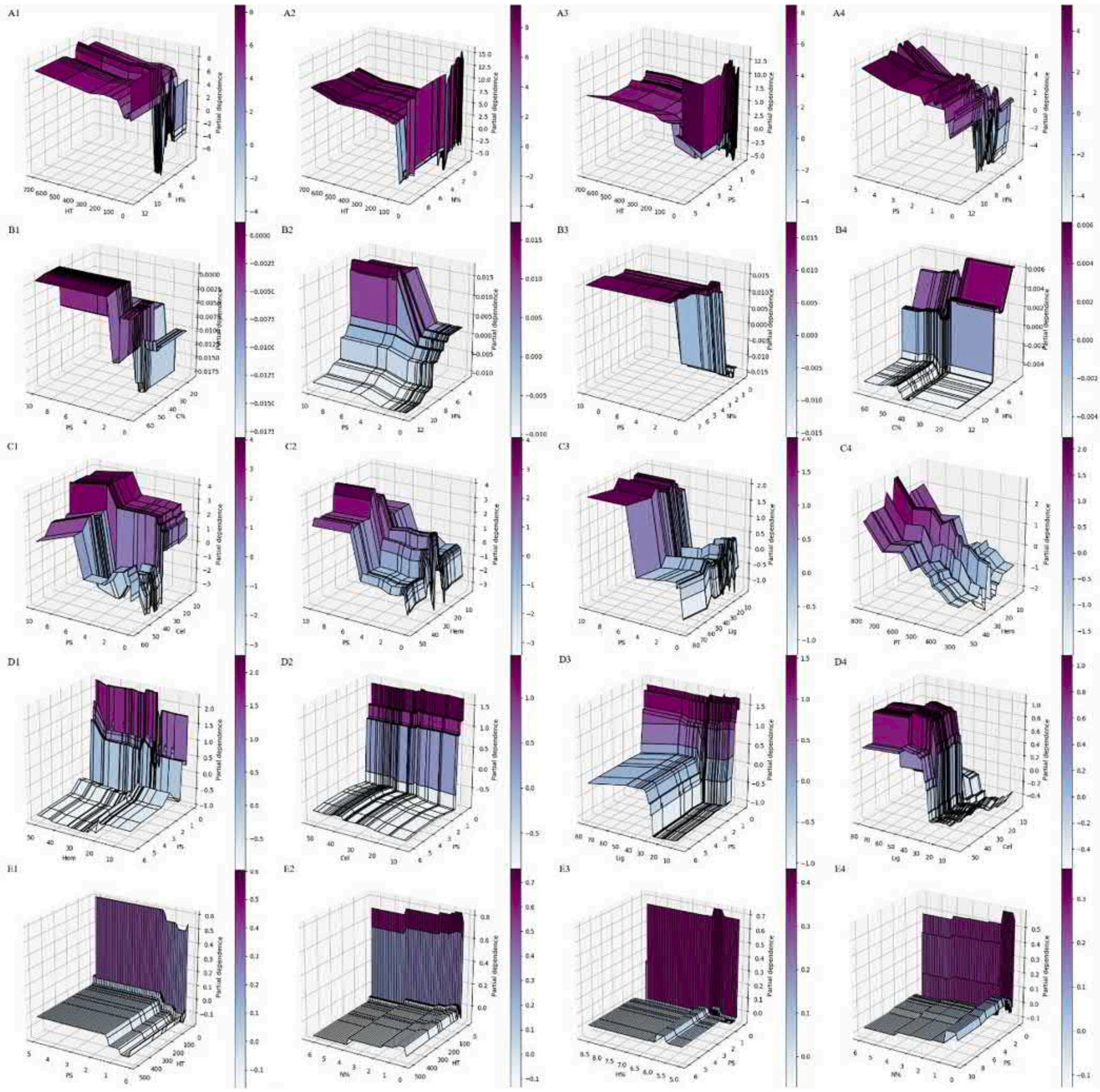

Fig. 9. The partial dependence plot of A-E represents the yield, viscosity, calorific value, H/C and O/C on the interactions between any two input variables.

lignin content (Fig. 9D4). Previous experiments also shows that the interaction between hemi-cellulose and lignin will be beneficial for the formation of phenolic compounds derived from lignin, while hinders the formation of hydrocarbons [57].

It is shown that biomass HT plays an obvious effect on the O/C, and PS and nitrogen content have limited influence compared with HT (Fig. 9E1 and E2). Compared with the original element content of biomass, the operating conditions have a more significant impact on O/C, while the effects of hydrogen and nitrogen content on O/C of bio-oil are relatively narrow (Fig. 9E3 and E4). The components in bio-oil is closely related to the compositions of raw materials, so for biomass with different element contents, the pyrolysis operation conditions become very important [58].

The two-factor PDP is a new route to study the relationship between raw material feedstock and pyrolytic parameters, which is helpful to analyze and predict the characteristics of bio-oil. Meanwhile, the reaction mechanism of raw materials can be understood by PDP with two-factor analysis, which is beneficial to the process optimization in engineering applications.

4. Conclusions

In this work, the machine learning method was used to comprehensively predict bio-oil from biomass pyrolysis, including quantity (yield), quality (viscosity and calorific value) and compositions (H/C and O/C) prediction of bio-oil. PCCs analysis could not completely extract the linear relationship between the yield, viscosity, calorific value, H/C, O/C and the variables of bio-oil. It can only simply analyze the quantity, quality and compositions index of bio-oil. The regression of viscosity, calorific value, H/C and O/C of bio-oil was realized by machine learning for the first time. The machine learning method of random forest can effectively regress the yield, viscosity, calorific value, H/C and O/C of bio-oil, and predict the relationship between target and feature through PDP analysis, and then visually interpret the complex relationship between target and feature. The random forest result demonstrated that the ultimate analysis data of biomass feedstock made a notable contribution to the yield, viscosity and O/C, while the chemical compositions data of biomass feedstock made greater contribution in the calorific value and the H/C of bio-oil.

CRediT authorship contribution statement

Tonghuan Zhang: Investigation, Methodology, Conceptualization, Writing – original draft. **Danyang Cao:** Data curation, Methodology. **Xin Feng:** Resources, Supervision. **Jiahua Zhu:** Resources. **Xiaohua Lu:** Resources. **Liwen Mu:** Conceptualization, Resources, Supervision, Writing – review & editing. **Hongliang Qian:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2021.122812.

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